

CHANDIGARH UNIVERSITY

University Institute of Sciences • Department of Mathematics
Master of Science in Data Science

DISSERTATION RESEARCH PROPOSAL

Explainable Machine Learning Models for Early Prediction of Heart Disease Using Clinical Data

Scoping the Contribution: A Verified Literature Screen and a Feasibility-Tested Framework for Cross-Cohort Distribution Shift, Calibration Decay and Explanation Stability

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Prepared following a systematic screen of 12 primary studies and an empirical feasibility test on the UCI Heart Disease cohorts.

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1. Executive Verdict

This proposal was written only after two verification steps were completed. First, every one of the twelve papers held in the reference folder was converted to text and screened automatically against nineteen search patterns covering concept drift, covariate shift, label shift, distribution shift, drift detection, model degradation, recalibration, retraining, continual and online learning, external validation, cross-site transfer, population-stability measures, calibration, Brier score and Expected Calibration Error, decision-curve analysis, SHAP, LIME and explanation stability. Second, the four UCI Heart Disease cohorts were loaded and the shift between them was measured directly, so that feasibility claims rest on computed statistics rather than on assumption.

Verdict 1. Concept drift is NOT novel in cardiovascular machine learning as a whole. Performance drift in cardiac-surgery risk models and data drift in emergency-department admission models are already published, and a 2025 systematic review covers explainability within concept and data drift. Any claim of outright novelty would not survive examination.

Verdict 2. Concept drift IS absent from the specific literature this dissertation sits in. Across all twelve screened papers there are zero occurrences of concept drift, covariate shift, label shift, distribution shift, drift detection, model degradation, recalibration, population-stability measures, decision-curve analysis or explanation stability. The gap within the UCI-benchmark heart-disease XAI line is real and documented.

Verdict 3. True temporal concept drift CANNOT be studied with the UCI heart-disease data. All four cohorts were collected in a single 1988 donation and carry no admission dates, no follow-up and no time index. Calling cross-cohort differences "concept drift" would be a scientifically invalid claim and is the single most likely reason an examiner would reject the work.

Verdict 4. Cross-cohort distribution shift IS defensible, and the shift is large enough to carry a dissertation. Measured against Cleveland, maximum-heart-rate shows a Population Stability Index of 1.88 at the VA cohort and 1.65 at Switzerland, against a conventional "major shift" threshold of 0.25; disease prevalence moves from 45.9% at Cleveland to 93.5% at Switzerland. These are not marginal effects.

The recommendation that follows from this is therefore not to abandon the idea, but to rename it precisely and to redesign the experiment around what the data can actually support. The study should be presented as an investigation of cross-cohort distribution shift, calibration decay and explanation stability under external validation, with adaptation by recalibration and retraining. Framed that way it is honest, it is feasible within an M.Sc. timeline, and it corrects a methodological weakness that runs through the entire body of work screened here.

2. What the Screened Literature Actually Contains

2.1 Method of the screen

The twelve PDF files were converted to plain text and searched with regular expressions rather than read selectively, so that absence of a concept could be asserted with confidence rather than inferred from memory. One file in the folder was an exact duplicate of another and was excluded, leaving twelve distinct studies. Counts below are occurrences of the pattern anywhere in the full text, including the reference list; a count of zero therefore means the concept is absent from the paper entirely.

2.2 The twelve studies

#	Study (first author, venue, year)	Data used	XAI	Calibration reported	External validation
1	Mohan et al., IEEE Access, 2019 — Hybrid ML techniques	UCI Cleveland (303)	No	No	No
2	Jindal et al., IOP Conf. Ser. MSE, 2021	UCI Cleveland (303)	No	No	No
3	Sanchula, M.Sc. Thesis, CSU San Bernardino, 2025 — XAI for HD prediction	UCI / Kaggle (303)	SHAP + LIME	No	No
4	Laftah & Al-Saedi, J. Robotics & Control, 2024 — Explainable ensembles	UCI Cleveland / Kaggle	SHAP + LIME	No	Named as future work
5	Wang et al., Comput. Biol. Med., 2021 — 3-year mortality in HF/CHD	Two Shanxi hospitals, China	SHAP	Yes — Brier score	Absent; named as a limitation
6	Chakraborty et al., Intelligence-Based Medicine, 2026 — Bangladesh cohort	Own Bangladesh cohort + UCI Cleveland	SHAP	No	Yes — one direction
7	Bhatt et al., Algorithms, 2023 — Effective HD prediction	Kaggle / UCI combined	No	No	No; single-dataset limitation stated
8	Dritsas & Trigka, Computers, 2024 — Deep learning + XAI	UCI-derived heart attack set	SHAP + LIME	No	No
9	Subramani et al., Front. Med., 2023 — ML with DL	UCI combined (918)	SHAP	Partial — Brier, Hosmer-Lemeshow	No
10	Krittanawong et al., Sci. Rep., 2020 — Meta-analysis of ML in CVD	344 screened studies	No	Discussed as a field-wide failing	Used as inclusion criterion
11	El-Sofany et al., Sci. Rep., 2024 — ML + XAI technique	CHDD + a private dataset, pooled	SHAP + LIME	No	Named as future work
12	Ganie et al., Sci. Rep., 2025 — Ensembles on multiple datasets	Two heart-disease datasets, trained separately	SHAP	No	No cross-dataset transfer

Table 1. The twelve screened studies. Only one reports a proper calibration metric as a primary result; only one performs a genuine cross-dataset transfer.

2.3 Screening matrix — the concepts that are missing

Concept searched	Papers containing it	Status
Concept drift	0 of 12	Entirely absent
Covariate shift	0 of 12	Entirely absent
Label / prior-probability shift	0 of 12	Entirely absent
Dataset / distribution / domain shift	0 of 12	Entirely absent
Drift detection (ADWIN, DDM, Page-Hinkley, KSWIN)	0 of 12	Entirely absent
Model degradation or performance decay	0 of 12	Entirely absent
Recalibration	0 of 12	Entirely absent
Retraining or model updating	1 of 12 (as future work only)	Named, never performed
Continual / online / incremental learning	2 of 12 (passing mention)	Never implemented
PSI, Kolmogorov-Smirnov, Wasserstein, Jensen-Shannon	0 of 12	Entirely absent
Decision-curve analysis / net benefit	0 of 12	Entirely absent
Explanation stability or rank agreement	0 of 12	Entirely absent
Brier score or Expected Calibration Error	2 of 12	Rare
External validation actually performed	1 of 12	Rare; three more name it as future work
SHAP	8 of 12	Standard practice
LIME	5 of 12	Common

Table 2. Automated keyword screen across the full text of all twelve studies. The first twelve rows define the research space that is open.

2.4 The nearest competitor, and why a gap remains

Chakraborty and colleagues, writing in *Intelligence-Based Medicine* in 2026, come closest. They train on an original Bangladeshi clinical cohort, apply SHAP, and externally validate against UCI Cleveland. That study is the correct benchmark to position against, and it should be cited as the closest prior work rather than ignored. Three things nevertheless remain undone in it. The transfer is performed in one direction only, so no claim can be made about which direction of shift is harmful. No calibration metric is reported at all, so the study cannot say whether the transferred probabilities remain usable for decision-making. And the SHAP analysis is descriptive: explanations are shown, but their stability across the two cohorts is never measured.

Most usefully, the authors state their own unfinished business explicitly: *"adaptive model updating strategies will be explored to ensure long-term performance stability as population characteristics evolve"* and they close by naming "multi-center prospective validation and adaptive model updating" as future work. A dissertation that measures shift, quantifies the resulting calibration and explanation decay, and tests whether recalibration recovers reliability is a direct and well-motivated response to a gap the prior literature has itself identified.

3. Feasibility: Which Question the Data Can Actually Answer

3.1 Why temporal concept drift is not available

Concept drift, in its strict definition, is a change in the conditional distribution of the outcome given the features, observed over time within a population. Studying it requires a time index. The four UCI Heart Disease cohorts were donated together in 1988 and contain no admission date, no collection period, no follow-up window and no ordering variable of any kind. There is no legitimate way to construct a temporal axis from them. Partitioning by age is sometimes proposed as a substitute; it is not one, because age is a predictor in the model and splitting on it induces the very covariate shift the study claims to have discovered. Any attempt to describe differences between the four cohorts as drift over time would be a category error, and would be visible as one to any examiner familiar with the dataset.

3.2 What the four cohorts do support — measured, not assumed

The four cohorts were loaded directly and compared. Cholesterol values recorded as zero were treated as missing, which is the documented UCI convention and materially changes the picture. The profile below is the empirical basis for the rest of this proposal.

Cohort	n	Prevalence	Mean age	Male	chol missing	ca missing	thal missing	slope missing
Cleveland (USA)	303	45.9%	54.4	68.0%	0.0%	1.3%	0.7%	0.0%
Hungarian (Budapest)	294	36.1%	47.8	72.4%	7.8%	99.0%	90.5%	64.6%
Switzerland (Zurich)	123	93.5%	55.3	91.9%	100.0%	95.9%	42.3%	13.8%
VA (Long Beach)	200	74.5%	59.4	97.0%	28.0%	99.0%	83.0%	51.0%

Table 3. Cohort profile computed from the raw UCI files. Prevalence ranges from 36% to 94% across sites; three of the thirteen features are effectively unavailable outside Cleveland.

3.3 Measured shift relative to Cleveland

Population Stability Index, the two-sample Kolmogorov–Smirnov statistic and the standardised Wasserstein distance were computed for the five continuous predictors, using Cleveland as the reference cohort. By the conventional reading of PSI, a value below 0.10 indicates a stable variable, 0.10 to 0.25 a moderate shift and above 0.25 a major shift.

Target cohort	Feature	PSI	KS D	KS p-value	Wasserstein / SD	Reading
Hungarian	age	0.956	0.355	2.8×10^{-17}	0.732	Major
	oldpeak	0.877	0.316	1.0×10^{-13}	0.391	Major
	thalach	0.341	0.261	1.9×10^{-9}	0.461	Major
	trestbps	0.296	0.062	0.58	0.081	Major (PSI) / n.s. (KS)
	chol	0.032	0.059	0.67	0.188	Stable
Switzerland	thalach	1.648	0.464	1.6×10^{-17}	1.226	Extreme
	trestbps	0.495	0.159	0.022	0.278	Major
	oldpeak	0.183	0.126	0.12	0.332	Moderate
	age	0.131	0.084	0.54	0.141	Moderate
	chol	—	—	—	—	Unmeasurable (100% missing)

Target cohort	Feature	PSI	KS D	KS p-value	Wasserstein / SD	Reading
VA	thalach	1.880	0.516	1.2×10^{-24}	1.172	Extreme
	age	0.405	0.262	9.0×10^{-8}	0.543	Major
	oldpeak	0.263	0.224	8.8×10^{-5}	0.297	Major
	chol	0.090	0.124	0.089	0.159	Stable
	trestbps	0.063	0.102	0.24	0.202	Stable

Table 4. Covariate shift measured against the Cleveland reference cohort. Maximum heart rate is the dominant shifted variable at both Switzerland and the VA.

3.4 Measured label shift

Comparison	Prevalence change	χ^2 statistic	p-value	Interpretation
Cleveland → Hungarian	45.9% → 36.1%	5.5	0.019	Significant prior shift
Cleveland → Switzerland	45.9% → 93.5%	80.4	3.0×10^{-19}	Severe prior shift
Cleveland → VA	45.9% → 74.5%	39.2	3.9×10^{-10}	Severe prior shift

Table 5. Label shift between cohorts. Every transfer direction carries a statistically significant change in outcome prevalence.

Why this matters for the field. Because prevalence moves from 46% to 94%, accuracy and F1 become severely misleading on transfer. A model that predicted disease for every Swiss patient would score 93.5% accuracy. Any transfer result in this literature reported as accuracy — and most are — is uninterpretable without the accompanying prevalence. Demonstrating this concretely is itself a contribution.

3.5 The feature-availability constraint

Table 3 shows that number of major vessels and thallium scan result are missing for 96% to 99% of patients outside Cleveland, and the exercise ST slope for 51% to 65% at two sites. Imputing a variable that is 99% absent does not recover information; it manufactures it, and it guarantees that the imputed value acts as a site marker. The transferable feature set is therefore the eleven predictors common to all four cohorts, and the dissertation must train the transfer model on those eleven. A thirteen-feature model may be reported separately as a Cleveland-only reference, but it cannot participate in the transfer experiment.

A trap to avoid. When all-zero cholesterol at Switzerland is replaced by a single pooled median, every Swiss record acquires an identical cholesterol value. That constant then identifies the site perfectly, and because the site has 93.5% prevalence, any pooled model can exploit it as a shortcut. Missingness must be handled per-cohort, and missingness itself should be recorded as a shift dimension rather than silently repaired.

4. Dataset Strategy

Three options were considered. The recommendation is Option A as the primary study, with Option B retained as an optional extension only if time allows.

Option	Data	What it supports	Access cost	Verdict
A — Primary	UCI Heart Disease: Cleveland 303, Hungarian 294, Switzerland 123, VA 200 (N = 920)	Cross-cohort distribution shift, label shift, calibration decay, explanation stability, recalibration and retraining. Not temporal drift.	None; already held	Recommended. Feasible now, shift verified as large.
B — Optional extension	NHANES continuous survey cycles with public mortality linkage	Genuine temporal shift across survey cycles, because cycles are dated. Outcome definition differs from the UCI label and must be redefined.	Public download; substantial data-preparation time	Strong if time permits. Verify cycle availability and outcome linkage before committing.
C — Not recommended	MIMIC-IV / eICU	True temporal validation; published cardiac work already does this	PhysioNet credentialing plus human-subjects training; multi-week delay	Realistic risk of consuming the dissertation timeline on access and cohort extraction alone.

Table 6. Dataset options assessed against what each can honestly support and what it costs to obtain.

4.1 Can the four UCI cohorts be used for shift experiments?

Yes, with three conditions attached, and those conditions should appear in the methods chapter rather than being buried in the limitations. First, the cohorts differ by site, by referral pattern and by measurement protocol, not by time, so the phenomenon under study is site heterogeneity and must be named as such. Second, the shift is confounded: the Swiss cohort is simultaneously more male, more diseased and missing cholesterol entirely, so no single cause can be isolated from a sample of four sites. Third, three sites have fewer than 300 patients, which means confidence intervals will be wide and every comparison must be reported with them.

4.2 The methodological limitations of treating separate datasets as drift

1. There is no temporal ordering, so nothing about degradation over time can be inferred. Cross-sectional differences between sites are not a time series.
2. With four sites, site is confounded with case mix, recruitment policy, equipment and coding practice. The study can quantify how much performance is lost, but not why, and should not speculate beyond what it measures.
3. Differential missingness is itself a shift. If it is imputed away before measurement, the study destroys the effect it set out to observe.
4. Outcome definition is only nominally identical. All four sites binarise a 0–4 angiographic severity code, but the referral threshold that brings a patient to angiography differs by site, which is a plausible source of genuine conditional shift.
5. Small target cohorts make the difference between two AUC values statistically fragile; a swing of five points may not be distinguishable from noise at $n = 123$.

4.3 How to simulate shift without making invalid claims

Controlled simulation should complement, never replace, the real cross-cohort experiment, and it should be presented in a clearly separate section. Three simulations are defensible and sufficient. Prevalence can be resampled within the Cleveland cohort to produce a sequence of target sets at 30%, 50%, 70% and 90% prevalence, isolating pure label shift with the conditional distribution held fixed by construction. Covariate shift can be induced by importance-resampling Cleveland patients with probability weighted by a single predictor such as age, isolating pure covariate shift. Conditional shift can be induced by flipping labels for a defined subgroup, for example reversing the outcome for a fraction of female patients with atypical chest pain. Because the ground truth is known by construction in all three cases, these simulations validate that the detection machinery works before it is applied to the real cohorts, where ground truth is unknown. They must be labelled as synthetic throughout, and no clinical conclusion may be drawn from them.

5. The Measurement Framework

The instruction to select the smallest defensible methodology has been taken seriously. What follows omits techniques that would add pages without adding evidence: no drift-detector zoo, no ten-model tournament, no deep learning, no federated learning. Three model families, four cohorts, and one metric per construct.

5.1 Separating covariate, label and conditional shift

The three forms of shift are distinguished by a sequence of tests, not by a single statistic. The logic is subtractive: measure the input shift, correct for it, measure the prior shift, correct for it, and attribute whatever degradation survives both corrections to genuine conditional shift.

Shift type	Formal statement	Primary test	Decisive evidence
Covariate shift	$P(X)$ differs; $P(Y X)$ unchanged	Per-feature PSI, two-sample KS with Benjamini–Hochberg correction across the eleven features; plus a domain classifier trained to separate source from target, reported as AUC	Domain-classifier AUC clearly above 0.5, with per-feature PSI above 0.25 identifying which variables carry it
Label shift	$P(Y)$ differs; $P(X Y)$ unchanged	χ^2 test on prevalence; prior correction by adjusting the model intercept to the target base rate	Prevalence change significant, and discrimination preserved after prior correction
Conditional (concept) shift	$P(Y X)$ differs	Residual degradation after both importance weighting for covariate shift and prior correction; corroborated by refitting the same model family on the target and comparing the decision function	Discrimination remains depressed after both corrections, and a target-refit model substantially outperforms the corrected transferred model

Table 7. The decomposition. Note that only the third row can support a claim about the conditional distribution, and only after the first two corrections have been applied.

The one sentence that protects the thesis. Discrimination measured by AUC is invariant to prevalence, whereas calibration, precision, F1 and accuracy are not. Reporting AUC alongside calibration slope and intercept therefore separates "the model ranks patients differently here" from "the model's probabilities mean something different here" — and those are two distinct findings that this literature currently conflates.

5.2 Metrics, by construct

Construct	Metrics retained	Deliberately excluded	Inference
Discrimination	ROC-AUC; PR-AUC; sensitivity and specificity at a threshold fixed on the source, and again at a threshold re-optimised on the target	Accuracy and F1 as headline measures — both are prevalence-dependent and misleading under the label shift documented in Table 5	DeLong test for paired AUC; 2,000-sample bootstrap percentile intervals for everything else
Calibration	Brier score; Expected Calibration Error with 10 uniform bins; calibration intercept; calibration slope; reliability diagram	Hosmer–Lemeshow, which is sensitive to sample size and gives no direction of miscalibration	Bootstrap intervals on slope and intercept; slope of 1 and intercept of 0 are the reference values
Clinical utility	Decision-curve analysis and net benefit across a clinically relevant threshold	Cost-weighted utility models requiring cost parameters that	Net benefit reported with bootstrap bands; no formal test, read

Construct	Metrics retained	Deliberately excluded	Inference
	range, compared against treat-all and treat-none	cannot be justified from this data	descriptively
Explanation stability	Global mean SHAP ranking compared by Spearman ρ and Kendall τ ; top-5 Jaccard overlap; sign agreement of mean SHAP per feature	LIME as a primary instrument — it is stochastic and its own run-to-run variance would confound the shift signal. Retain LIME only as a secondary check on the top-3 features	Bootstrap intervals on ρ , τ and Jaccard; run-to-run variance quantified first so that instability is not mistaken for shift
Distribution shift	PSI per feature; two-sample KS with BH correction; standardised Wasserstein distance; domain-classifier AUC as the single multivariate summary	ADWIN, DDM, Page-Hinkley, KSWIN and the rest of the streaming-detector family — all require a data stream, which does not exist here	Permutation null for the domain-classifier AUC; BH-adjusted p-values across the eleven features
Adaptation	Recalibration-in-the-large (intercept only); logistic recalibration (intercept and slope); full retraining on the target; before-and-after comparison on every metric above	Continual and online learning, transfer learning architectures, domain-adversarial training — none is justifiable at these sample sizes	Paired comparison on a held-out target partition never used for adaptation

Table 8. The retained measurement set, with the exclusions stated explicitly. The exclusions are as much a part of the methodological argument as the inclusions.

5.3 Measuring explanation stability honestly

This is the part of the design most likely to be challenged, and it needs one precaution that the existing literature does not take. If SHAP values are computed on the source cohort and again on the target cohort, the rankings will differ for two entirely different reasons: because the model behaves differently, or simply because the target patients have different feature values. Only the first is a finding. The precaution is to compute explanations for both models on a common evaluation set, so that the data are held constant and only the model varies. The study should therefore report three quantities: the run-to-run variance of the explainer itself, which establishes the noise floor; the ranking divergence attributable to the change in patient mix; and the ranking divergence attributable to the change in model. Only the third supports a claim about explanation instability under shift.

6. Experimental Protocol

6.1 Design

The primary design is leave-one-site-out. Each of the four cohorts is held out in turn as the external target while the remaining three form the development set. This yields four transfer directions rather than one, which is what allows the study to say something about the direction and magnitude of shift rather than reporting a single anecdote. Hyperparameters are selected by nested cross-validation inside the development set only. The held-out site is never seen during development, and is itself split so that adaptation and post-adaptation evaluation use disjoint patients.

6.2 Models

Three model families, chosen for contrast rather than for count. Penalised logistic regression serves as the interpretable reference and is the model for which calibration slope and intercept have their clearest meaning. Random forest represents bagged trees. Gradient-boosted trees, in one implementation only, represent the current benchmark. Ten-model tournaments of the kind common in this literature add comparison rows without adding evidence, and they inflate the winner's-curse problem when the best cross-validation score is also reported as the honest estimate.

6.3 Stage-by-stage specification

Stage	Operation	Measured and recorded at this stage
1. Cohort assembly	Load four cohorts separately. Recode UCI missing markers, including cholesterol zero. Retain the eleven common features. Record cohort membership as metadata, never as a predictor.	n per site; prevalence; per-feature missingness; the cohort profile of Table 3
2. Leakage-safe preprocessing	Split first, then fit. Imputers and scalers are fitted on development data only and applied unchanged to the target. Imputation is fitted per fold, never on the pooled data.	A written preprocessing audit confirming that no target-derived statistic enters any fitted object
3. Baseline development	Nested cross-validation on the development sites; outer folds give the honest internal estimate, inner folds select hyperparameters.	Internal ROC-AUC, PR-AUC, Brier, ECE, calibration slope and intercept, each with bootstrap intervals
4. Source explanations	SHAP on the development set; establish the explainer noise floor by repeated runs.	Global mean SHAP ranking; run-to-run variance of that ranking
5. Shift detection	Compare development against held-out target before any prediction is made.	PSI, KS with BH correction, standardised Wasserstein per feature; domain-classifier AUC with permutation null; prevalence χ^2
6. Transfer evaluation	Apply the frozen model to the held-out site. Do not retune, do not re-threshold.	ROC-AUC, PR-AUC; sensitivity and specificity at the source-fixed threshold; Δ AUC with DeLong test
7. Calibration decay	Assess the frozen model's probabilities on the target.	Brier, ECE, calibration intercept, calibration slope, reliability diagram — the core degradation result
8. Clinical utility decay	Decision-curve analysis on the target for the frozen model.	Net benefit against treat-all and treat-none across the threshold range
9. Explanation instability	Compute explanations for the frozen model and for a target-refit model on a common evaluation set.	Spearman ρ , Kendall τ , top-5 Jaccard, sign agreement — each separated into mix-driven and model-driven components
10. Shift attribution	Apply importance weighting for covariate shift,	Residual degradation after each correction; the covariate /

Stage	Operation	Measured and recorded at this stage
	then prior correction for label shift.	label / conditional decomposition of Table 7
11. Recalibration	Fit intercept-only and then intercept-and-slope recalibration on a target adaptation partition.	All calibration metrics before and after; discrimination before and after, to test whether it is unchanged
12. Retraining	Refit on the target adaptation partition; also fit a pooled model with site included, as a third comparator.	All metrics on the untouched target evaluation partition; sample size needed to recover performance
13. Synthesis	Aggregate across the four leave-one-site-out folds.	Which degradation recalibration recovers, which it does not, and what that implies for deployment

Table 9. The protocol. Stages 7, 9 and 11 carry the novel contribution; stages 1, 2 and 6 carry its credibility.

The expected headline result. Recalibration is expected to restore calibration slope toward 1 and reduce ECE substantially, while leaving ROC-AUC essentially unchanged. Retraining is expected to be the only intervention that recovers discrimination, and to require more target data than a small site can supply. If that pattern holds across the four folds it is a clean, quotable and clinically meaningful finding — and it is a finding that none of the twelve screened papers is positioned to report.

7. The Proposal

A. Research title

Recommended title. Explainable Machine Learning for Early Prediction of Heart Disease: Quantifying Cross-Cohort Distribution Shift, Calibration Decay and Explanation Stability Under External Validation.

A shorter variant for the title page, if a single line is required: “Explainable Machine Learning Models for Early Prediction of Heart Disease: A Cross-Cohort Study of Distribution Shift, Calibration and Explanation Stability.” The word drift does not appear in either, deliberately. It may appear in the literature-review chapter, where the distinction between temporal drift and cross-sectional shift should be drawn explicitly — that discussion is itself a mark of methodological maturity.

B. Research gap statement

Research on machine learning for heart-disease prediction has converged on a narrow and repeatable template: a single benchmark cohort, an ensemble of classifiers, an accuracy figure above ninety per cent, and a SHAP plot. A screen of twelve representative studies confirms the pattern. Eight report SHAP and five report LIME, yet only two report any calibration metric, only one performs a genuine external validation, and not one reports decision-curve analysis, quantifies distribution shift, measures the stability of its own explanations, or tests whether a model can be repaired once it is moved. Because outcome prevalence across the publicly available cohorts ranges from thirty-six to ninety-four per cent, the accuracy figures on which the field rests are not comparable across settings, and the explanations offered to clinicians have never been shown to hold when the population changes. The gap is therefore not another algorithm; it is evidence about whether these models and their explanations survive contact with a different patient population, and whether anything can be done when they do not.

C. Research problem

Explainable heart-disease models are developed and reported under an implicit assumption that the deployment population matches the development population. That assumption is demonstrably false across the four cohorts in the standard benchmark, where maximum heart rate alone shows a Population Stability Index of 1.88 between Cleveland and the VA and prevalence more than doubles between Cleveland and Switzerland. The consequence is unknown: it is not established how much discrimination is lost on transfer, whether calibration fails faster than discrimination, whether the SHAP explanations that justify clinical trust remain stable, or whether lightweight adaptation can restore reliability without new large-scale data collection. This study addresses that problem.

D. Research questions

6. RQ1. How large is the distribution shift between the four UCI heart-disease cohorts, and how does it decompose into covariate shift, label shift and residual conditional shift?
7. RQ2. How much predictive performance is lost when an explainable model developed on three cohorts is applied unchanged to a held-out fourth, and does the loss differ by transfer direction?
8. RQ3. Does calibration degrade disproportionately relative to discrimination under cross-cohort transfer, and what does that imply for the accuracy-based reporting that dominates this literature?
9. RQ4. How stable are SHAP-based global explanations across cohorts, once the explainer's own run-to-run variance and the change in patient mix are separated from the change in model behaviour?
10. RQ5. Can logistic recalibration on a small target sample restore calibration and clinical net benefit, and does it also restore discrimination?

11. RQ6. What volume of target data is required before retraining outperforms recalibration, and is that volume realistic for a small clinical site?

E. Objectives

12. To assemble the four UCI cohorts under a leakage-safe, per-cohort preprocessing protocol that preserves missingness as an observable rather than repairing it away.
13. To quantify cross-cohort shift using PSI, Kolmogorov-Smirnov, standardised Wasserstein distance and a domain classifier, and to decompose it into covariate, label and conditional components.
14. To develop three model families under nested cross-validation and evaluate them internally on discrimination, calibration and clinical utility.
15. To measure degradation in discrimination, calibration and net benefit under leave-one-site-out external transfer.
16. To quantify the stability of SHAP explanations across cohorts, separating explainer noise and patient-mix effects from genuine model-behaviour change.
17. To evaluate recalibration and retraining as recovery strategies and to characterise what each does and does not restore.

F. Hypotheses

	Hypothesis	Test
H1	Cross-cohort transfer produces a statistically significant loss of discrimination in at least three of the four leave-one-site-out folds.	DeLong test on paired AUC; bootstrap interval on Δ AUC excluding zero
H2	Calibration degrades proportionally more than discrimination: calibration slope departs from 1 and intercept from 0 by a larger standardised margin than the relative fall in AUC.	Bootstrap intervals on slope and intercept; standardised comparison of relative degradation
H3	Accuracy and F1 systematically overstate transfer performance relative to AUC, PR-AUC and net benefit, and the overstatement scales with the magnitude of prevalence shift.	Correlation between prevalence change and the accuracy-minus-AUC discrepancy across folds
H4	Global SHAP rankings are unstable across cohorts beyond the explainer noise floor, with Kendall τ below 0.6 in at least one fold.	Bootstrap interval on τ compared against the measured run-to-run variance
H5	Logistic recalibration restores calibration and net benefit without materially restoring discrimination.	Paired before-and-after comparison on the held-out target partition
H6	Degradation persists in at least one cohort after both importance weighting and prior correction, indicating genuine conditional shift.	Residual Δ AUC after both corrections, with bootstrap interval

Table 10. Hypotheses and the test attached to each. Every hypothesis is falsifiable on the available data.

G. Experimental design

Retrospective secondary analysis of four independent clinical cohorts, with a leave-one-site-out external-validation design, nested cross-validation for model development, and a pre-specified analysis plan. The design is observational and comparative; no causal claim is made about why sites differ. Stage-by-stage detail is given in Table 9 and the full measurement specification in Table 8.

H. Dataset strategy

The UCI Heart Disease repository provides four independent cohorts totalling 920 patients: Cleveland (303), Hungarian (294), Switzerland (123) and Long Beach VA (200). Analysis proceeds on the eleven predictors

recorded at all four sites; number of major vessels and thallium scan result are excluded from the transfer experiment because they are missing for 96% to 99% of non-Cleveland patients, and a thirteen-feature Cleveland-only model is reported separately as a reference rather than as a comparator. Cholesterol values of zero are treated as missing throughout. Should time permit, NHANES survey cycles are retained as an optional extension providing a genuinely dated axis, subject to prior verification of cycle coverage and outcome linkage.

I. Statistical analysis plan

- Discrimination: ROC-AUC and PR-AUC, with 2,000-sample bootstrap percentile intervals; paired AUC comparisons by DeLong's test.
- Calibration: Brier score, Expected Calibration Error over 10 uniform bins, calibration intercept and calibration slope from a logistic regression of the outcome on the linear predictor, each with bootstrap intervals; reliability diagrams reported alongside.
- Clinical utility: decision-curve analysis with net benefit against treat-all and treat-none, with bootstrap bands, interpreted descriptively.
- Shift: per-feature PSI, two-sample Kolmogorov-Smirnov tests with Benjamini-Hochberg control of the false discovery rate across eleven features, standardised Wasserstein distance, and domain-classifier AUC assessed against a permutation null.
- Explanation stability: Spearman ρ , Kendall τ and top-5 Jaccard overlap between global mean |SHAP| rankings, with bootstrap intervals, benchmarked against the explainer's measured run-to-run variance.
- Adaptation: paired before-and-after comparison on a target evaluation partition disjoint from the adaptation partition; learning curves for retraining across increasing target sample sizes.
- Multiplicity: the false discovery rate is controlled within each family of tests. No result is reported as significant on an uncorrected p-value. Given three target cohorts of 123 to 294 patients, the study is explicitly powered to detect large effects only, and intervals are reported in preference to p-values throughout.

J. Expected contribution

18. The first study in the UCI heart-disease explainable-ML line to quantify cross-cohort distribution shift and decompose it into covariate, label and conditional components.
19. The first to report calibration slope, calibration intercept and decision-curve net benefit under external transfer in this literature, establishing whether transferred probabilities remain usable for clinical decisions.
20. A method for measuring explanation stability that separates explainer noise and patient-mix effects from genuine model-behaviour change — a distinction not drawn in any of the twelve screened studies.
21. Evidence on whether lightweight recalibration is sufficient to redeploy a heart-disease model at a new site, or whether retraining is unavoidable, with an estimate of the target sample size required.
22. A reusable, leakage-safe, leave-one-site-out benchmark protocol and codebase for the four public cohorts, released so that future work can report transfer results comparably.
23. A documented demonstration that accuracy-based reporting is not comparable across these cohorts, which has direct implications for how results in this field should be read.

K. Limitations

- The cohorts carry no time index, so nothing in this study speaks to temporal concept drift or to degradation over time. The word drift is avoided in the title and claims for that reason.

- Shift is confounded with site: with four cohorts it is impossible to separate case mix, referral threshold, measurement protocol and coding practice.
- Sample sizes are small — 123 at Switzerland, 200 at the VA — so confidence intervals will be wide and only large effects are detectable.
- The data were collected in 1988 and the clinical practice they reflect is historical; findings about transfer mechanics generalise, findings about particular risk factors do not.
- Cholesterol is entirely unrecorded at Switzerland, so one predictor cannot be assessed for shift at that site at all.
- The outcome is angiographic severity binarised at a threshold, which is a proxy for clinically significant coronary artery disease and is subject to site-specific referral bias.
- The study is retrospective and observational; no prospective or clinical-impact claim is made.

L. Fallback research design

If the shift-decomposition analysis proves too fragile at these sample sizes — specifically, if the residual degradation after importance weighting and prior correction cannot be estimated with a usable confidence interval — the study should contract rather than substitute. The fallback retains stages 1 through 9 and 11 of Table 9 and drops the formal decomposition and the retraining learning curves. That reduced study is: a rigorous leave-one-site-out external validation of explainable heart-disease models, reporting discrimination, calibration and clinical utility; a quantification of explanation stability under cohort change; and a test of whether recalibration restores reliability. Measured against the screened literature, where only one of twelve studies performs external validation at all and none reports calibration slope, net benefit or explanation stability, that reduced study is still a clear contribution and is achievable with certainty on the data already held. The fallback should be written into the proposal from the outset, not improvised later.

8. Immediate Next Steps

#	Action	Why it comes first
1	Rewrite the working title and abstract to remove the word "drift" and name cross-cohort distribution shift instead.	Protects the entire thesis from the single most likely line of examiner challenge.
2	Rebuild the data layer so the four cohorts load separately, cholesterol zeros are recoded as missing, and the eleven-feature transfer set is fixed.	Every downstream result depends on this; the pooled-median treatment currently creates a site marker.
3	Reproduce Tables 3, 4 and 5 as the first results section of the dissertation.	They are already computed, already defensible, and they establish the premise of the whole study.
4	Implement the leave-one-site-out harness with nested cross-validation and bootstrap intervals before adding any further model.	Design credibility is worth more than model count.
5	Cite Chakraborty et al. (2026) as the closest prior work and position against its three stated gaps.	Shows command of the field and pre-empts the question of what is new.
6	Add calibration slope, calibration intercept and decision-curve net benefit to the evaluation code.	These are the metrics that make the contribution distinctive, and they are absent from almost all prior work.

Table 11. Ordered so that the items protecting the thesis come before the items extending it.

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Additional works consulted for the drift and shift literature, to be verified and formatted on citation: a 2024 retrospective analysis of performance drift in machine-learning models for cardiac-surgery risk prediction (JMIRx Med); a 2025 systematic literature review of explainability and interpretability in concept and data drift (Algorithms); and published work using explainable machine learning to characterise data drift in emergency-department admissions.

Note on sources. The twelve primary studies above were screened in full text from the reference folder. Claims in Section 3 about cohort size, prevalence, missingness and shift magnitude were computed directly from the raw UCI data files and are reproducible from them. Claims about the wider drift literature in Section 1 derive from published titles and abstracts located during the screen and should be read in full before citation in the dissertation itself.